Decision Trees for Transparent Decision-Making

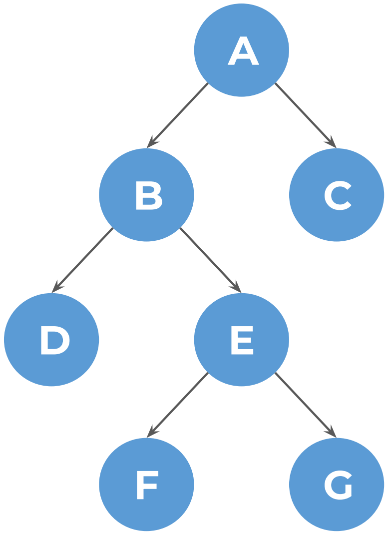
Agenda

- Introduction to Decision Tree
- Pruning Techniques: Pre-Pruning vs Post-Pruning
- Impurity Measures for Decision Trees (Classification)

Let's begin the discussion by answering a few questions

Classification Methods Quiz

Which of the following nodes are leaf nodes?



A. A, B, E

B. F, G

C. C, D, F, G

D. A, B, C

Classification Methods Quiz

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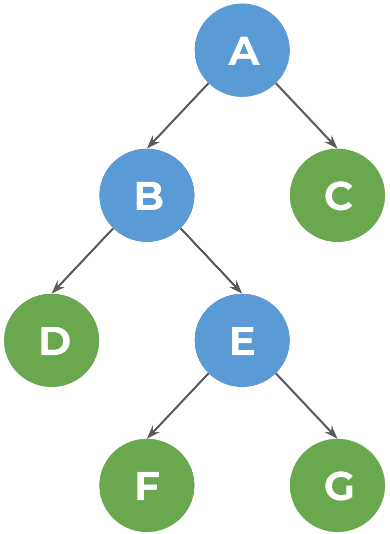
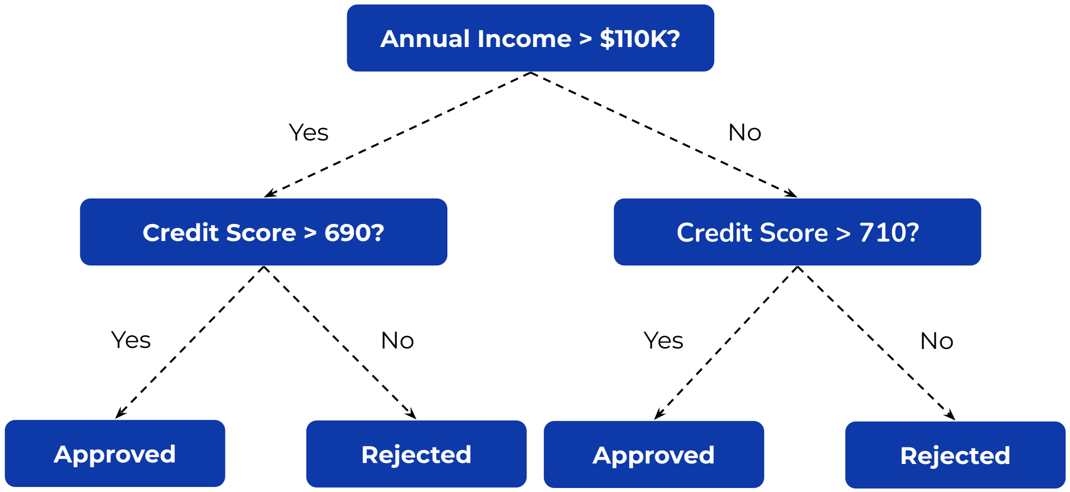


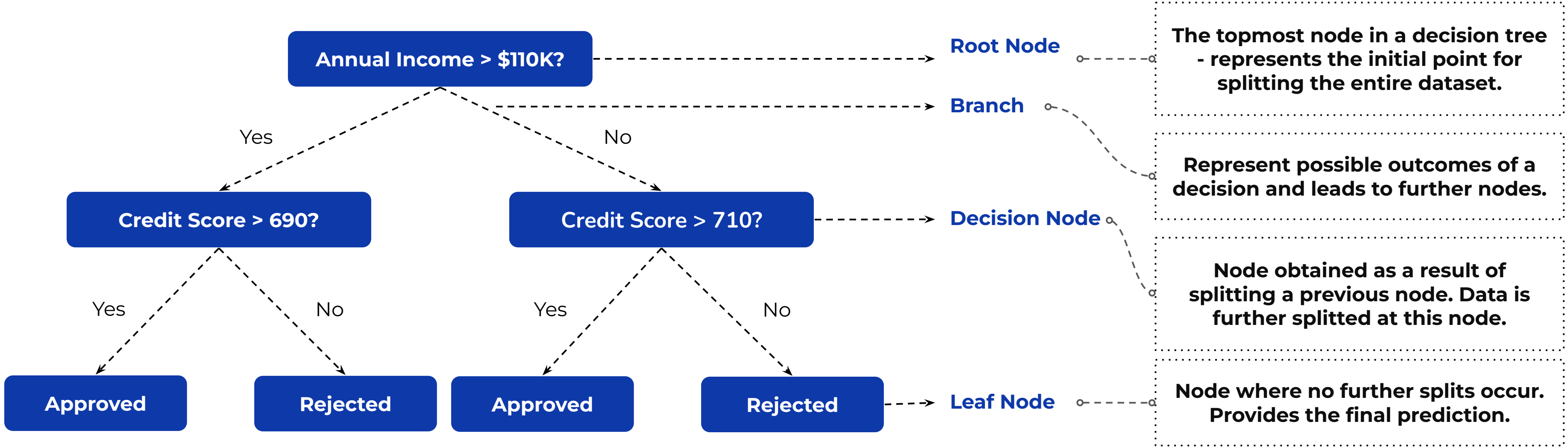
- A — A, B, E
- B — F, G
- C — C, D, F, G
- D — A, B, C

Introduction to Decision Tree

A **decision tree is** a popular and powerful supervised learning algorithm that works with both categorical and continuous variables by visually representing all possible decisions based on given conditions, where the training data is repeatedly split into smaller groups according to defined rules to achieve better prediction accuracy.



Decision Tree Structure



At the **leaf node**, find the **proportion of each class**

The **class** with the **higher proportion** is the **prediction**

Decision Tree Quiz

A decision tree works like a warehouse sorter. As each item (data point) arrives, it decides which chute to send it through. How does it actually handle different types of data during this sorting process?

A. Numerical → threshold splits; Categorical → group-based splits

B. Numerical → alphabetical splits ; Categorical → threshold splits

C. Numerical → threshold splits; Categorical → threshold splits

D. Numerical → group-based splits; Categorical → group-based splits

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How Decision Trees Handle Data

Decision Trees

Numerical Data

For each numerical feature, the algorithm identifies a **threshold**
$X_i \leq$ **threshold**.

The threshold **best splits the data into two groups** to **maximise information gain or minimise impurity** (**Gini Impurity** or **Entropy**).

Categorical Data

If the categorical feature has only **two categories** (e.g., "Yes" or "No"), the tree can **split the data** easily **based on the presence of one category**.

For features with **multiple categories** (e.g., "Red," "Blue," "Green"), the tree considers **all possible subsets of the categories**. This can involve **splitting into two groups by grouping some categories together** (e.g., {Red, Blue} vs. {Green}).

How Decision Trees Handle Data

Great Learning
POWER AHEAD

Decision Trees

- **Numerical Data**
  - Is $\mathbf{X_i} < \mathbf{0}$?
    - All **negative values** of X_i
    - All **positive values**(and zero) of X_i
- **Categorical Data**
  - Is $\mathbf{X_i}$ = [**Green**, **Blue**]?
    - [**Green**, **Blue**]
    - [**Red**]

Decision Tree Quiz

You are building a Decision Tree model for fraud detection with a large dataset and limited computational resources. To ensure faster training and prevent overfitting, which pruning strategy should you choose?

A. Post-Pruning

B. Pre-Pruning

C. Both Pre-Pruning and Post-Pruning

D. No pruning is required

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Pre-Pruning vs Post-Pruning

Decision trees can become too large and overfit the training data, so pruning simplifies the tree by removing unnecessary branches and converting some branch nodes into leaf nodes.

Pre-Pruning	Post-Pruning
Prevents the decision tree from **growing fully** by introducing a **stopping criterion**	Allows the **decision tree to grow fully** and then removes **unnecessary branches and/or leaves**
Prevents the tree from **capturing noise and overfitting** on the training data	The **Nodes and Subtrees** are **replaced** with **leaves** to **reduce complexity**
Faster and computationally efficient as the entire tree is not built	**Slower and computationally expensive** as it requires building the entire tree first before trimming unnecessary branches

Decision Tree Quiz

Which of the following parameters is not used for pre-pruning a Decision Tree?

- A. max_depth
- B. min_samples_leaf
- C. ccp_alpha
- D. min_samples_split

Decision Tree Quiz

Which of the following parameters is not used for pre-pruning a Decision Tree?

A max_depth

B min_samples_leaf

C ccp_alpha

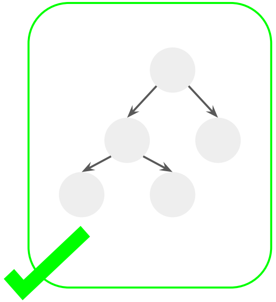
D min_samples_split

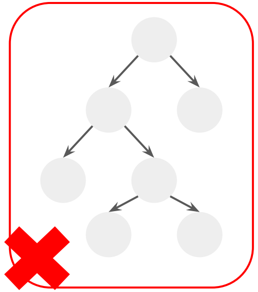
Pre-Pruning a Decision Tree

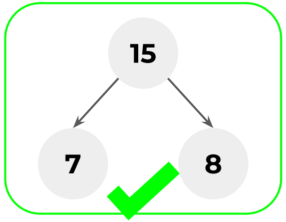
Great Learning
POWER AHEAD

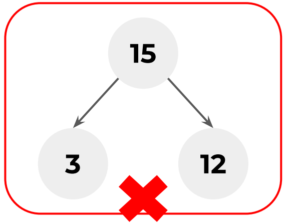
Hyperparameters that help pre-prune a decision tree

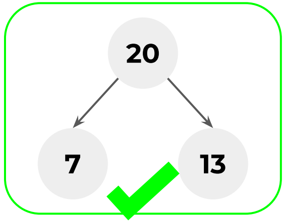
Max Depth	Min Samples for Leaf	Min Samples to Split
Sets the maximum depth to which a tree can grow	The minimum number of samples to be present in every leaf node	The minimum number of samples to be present in a node for it to be eligible for a split
max_depth = 2	min_samples_leaf = 5	min_samples _split = 20

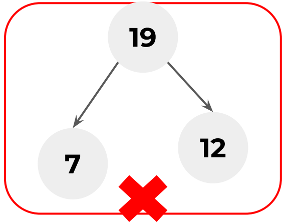












Post-Pruning a Decision Tree

Hyperparameter that help post-prune a decision tree

Cost Complexity Pruning

ccp_alpha

Balances tree complexity with classification accuracy by penalizing the number of nodes

Decision Tree Quiz

Which of the following attribute-threshold combinations should be chosen for the first split in a decision tree based on Gini Impurity reduction?

- Annual Income > $110K | Gini before split = 0.43, after split = 0.37
- Credit Score > 690 | Gini before split = 0.45, after split = 0.4
- Annual Income > $70K | Gini before split = 0.45, after split = 0.4
- Credit Score > 700 | Gini before split = 0.44, after split = 0.41

A. Credit Score > 690

B. Annual Income > $ 70k

C. Annual Income > $ 110k

D. Credit Score > 700

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A. Credit Score > 690

B. Annual Income > $ 70k

C. Annual Income > $ 110k

D. Credit Score > 700

Impurity Measures

Gini Impurity	Entropy
Measures the probability of misclassifying a randomly selected data point if we label it randomly	Measures how mixed up or organized data points are in a certain group.
Higher the Gini Impurity, lower the homogeneity - worse the split	Higher the Entropy, lower the homogeneity - worse the split
Ranges between 0 (pure) and 0.5 (impure)	Ranges between 0 (pure) and 1 (impure)

$$Gini\ Impurity = 1 - \sum_{i=1}^{k} p_i^2$$

$$Entropy = -\sum_{i=1}^{k} p_i \log_2(p_i)$$

p_i = proportion (probability) of class i in the node; k = total number of classes

Power Ahead !

Classification vs Regression Trees

Classification Trees	Regression Trees
Predict a categorical value (class label)	Predict a numerical value
Uses Impurity measures like Gini Impurity and Entropy for splitting	Uses measures like Mean Squared Error (MSE) for splitting
Focuses on reducing impurity with each split	Focuses on reducing variance with each split